Operational Research & Optimization (STA201)

2026 Spring Semester Midterm

1. (40 points) Answer the following questions.

   (a) (5 points) Given a vector norm, write down the definition of matrix norm.

   (b) (5 points) Write down the definition of the second-order cone program (SOCP).

   (c) (5 points) Prove that

   $$C = \left\{ x \in \mathbb{R}^n \,\middle|\, \|x - x_c\|_2 \leq r,\ x^T x_D > 0 \right\}$$

   is a convex set, where $r > 0$ and $x_c, x_D \in \mathbb{R}^n$. Assume that C is not a null set.

   (d) (5 points) Write down the separation hyperplane theorem.

   (e) (5 points) Given a basic solution

   $$x = \begin{bmatrix} x_B \\ 0 \end{bmatrix}$$

   from a linear system $Ax = b$, write down the definition of a degenerate basic solution.

   (f) (5 points) Let f be convex with a nonempty $\operatorname{dom} f$. Show that $\partial f(x)$ is the intersection of an infinite set of halfspaces.

   (g) (10 points) Formulate the following problem into the standard form and canonical form of the linear programming problem, respectively:

   $$\begin{aligned} \min \quad & |x_1| + 5x_2 \\ \text{subject to} \quad & x_1 + x_2 \geq 10, \\ & x_2 \geq 0. \end{aligned}$$

2. (14 points) Use the simplex method to solve the following LPP:

$$\begin{aligned} \min \quad & -5x_1 - 2x_2 - x_3 - x_4 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 + 2x_4 \leq 6, \\ & 3x_1 + x_3 \leq 15, \\ & 5x_1 + 4x_2 + x_4 \leq 24, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

3. (16 points) Consider an optimization problem

$$\min f(x) \quad \text{s.t. } x \in \Omega,$$

where f is a convex function and Ω is a convex set. Prove that any local minimum is also a global minimum.

4. (18 points) A function $f : \mathbb{R}^n \to \mathbb{R}$ is called quasiconvex if

$$f(\lambda x + (1-\lambda)y) \le \max\{f(x), f(y)\}, \qquad \forall x, y,\ \lambda \in [0,1].$$

Show that a function is quasiconvex if and only if the level set

$$\{x \mid f(x) \le a\}$$

is convex for all $a \in \mathbb{R}$.

5. (12 points) Consider the following problem in standard form:

$$\begin{aligned} \max \quad & x_0 = 2x_1 - x_4 - 5x_5 + 2x_7 \\ \text{subject to} \quad & x_1 + x_3 - 2x_5 - x_7 = 12, \\ & -x_1 + x_2 + x_4 + 3x_5 - x_7 = 6, \\ & 2x_1 - 2x_4 + 6x_5 + x_6 + 4x_7 = 18, \\ & x_j \ge 0, \qquad j = 1, \dots, 7. \end{aligned}$$

The tableau below represents the current basic solution:

x_1	x_2	x_3	x_4	x_5	x_6	x_7	b
0	3	1	1	b	1	0	f
0	1	0	0	c	$\frac{1}{2}$	1	g
1	1	1	0	d	$\frac{1}{2}$	0	h
0	1	1	0	a	1	0	e

(a) (4 points) Instead of using the simplex method, directly identify the basic variables and the basis inverse (the inverse matrix of the basic matrix) corresponding to the given tableau.

(b) (8 points) Determine the values of the unknowns a, b, c, d, e, f, g, h in the tableau. Write down each step of your calculation instead of only giving all the values.